Experiment 12 : Write a Program to Implement Random Forest Algorithm on Digit Dataset of Sklearn

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_digits
digits = load_digits()
```

In [2]:
```
dir(digits)
```

Out[2]:
```
['DESCR', 'data', 'feature_names', 'frame', 'images', 'target', 'target_names']
```

In [3]:
```
digits.data
```

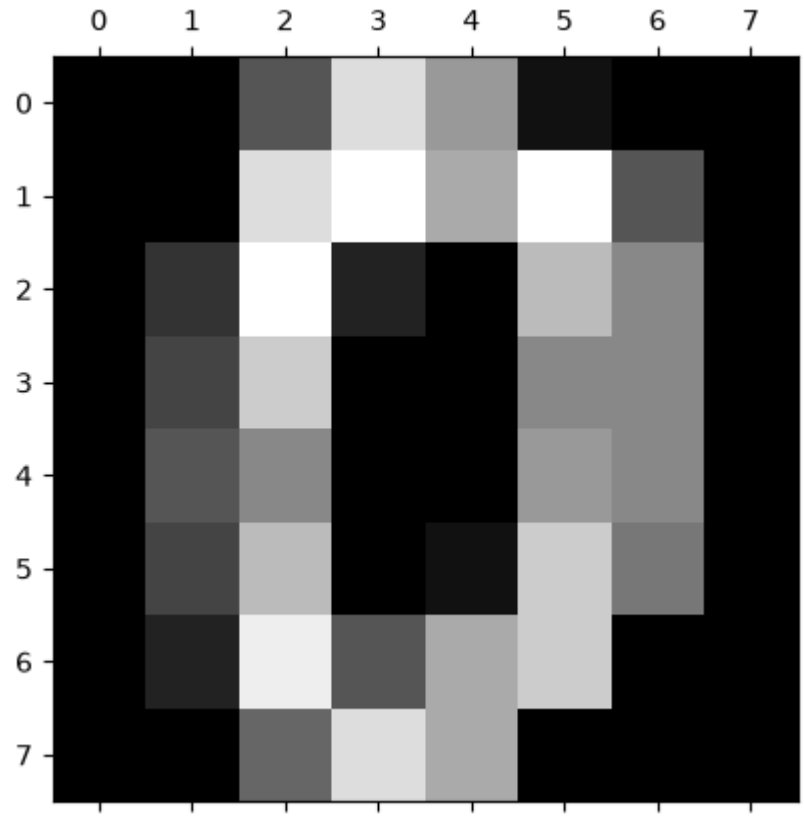
Out[3]:
```
array([[ 0.,  0.,  5., ...,  0.,  0.,  0.],
       [ 0.,  0.,  0., ..., 10.,  0.,  0.],
       [ 0.,  0.,  0., ..., 16.,  9.,  0.],
       ...,
       [ 0.,  0.,  1., ...,  6.,  0.,  0.],
       [ 0.,  0.,  2., ..., 12.,  0.,  0.],
       [ 0.,  0., 10., ..., 12.,  1.,  0.]], shape=(1797, 64))
```

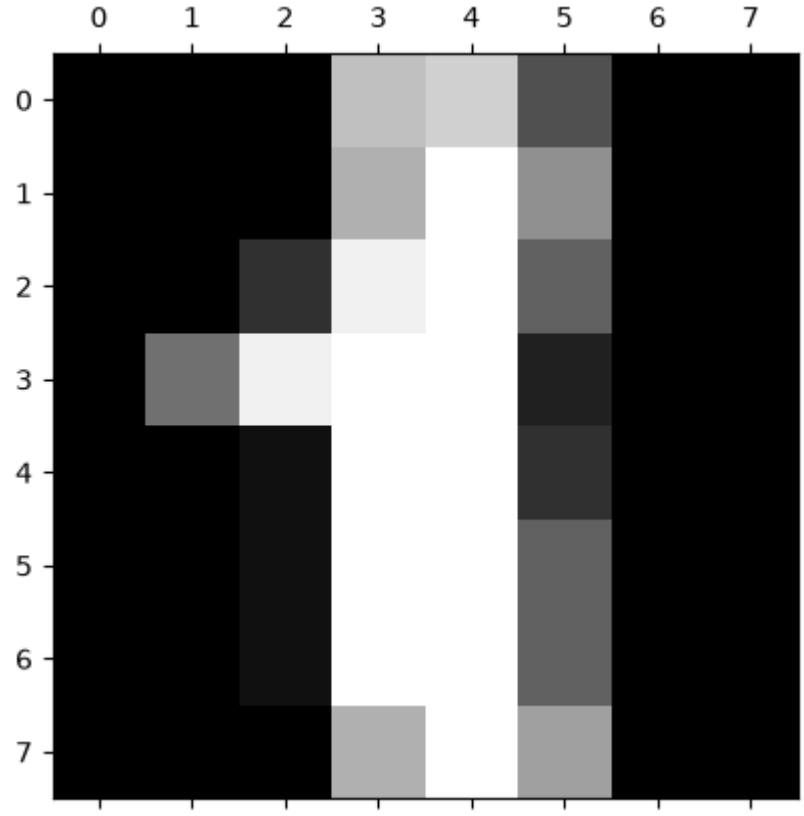
In [4]:
```
digits.target_names
```

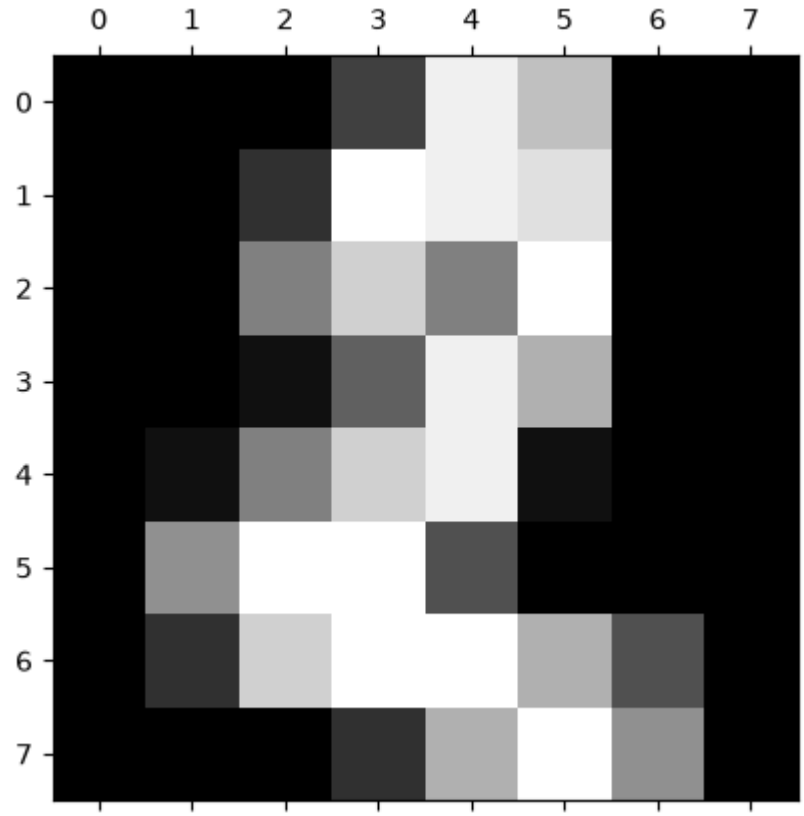
Out[4]:
```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

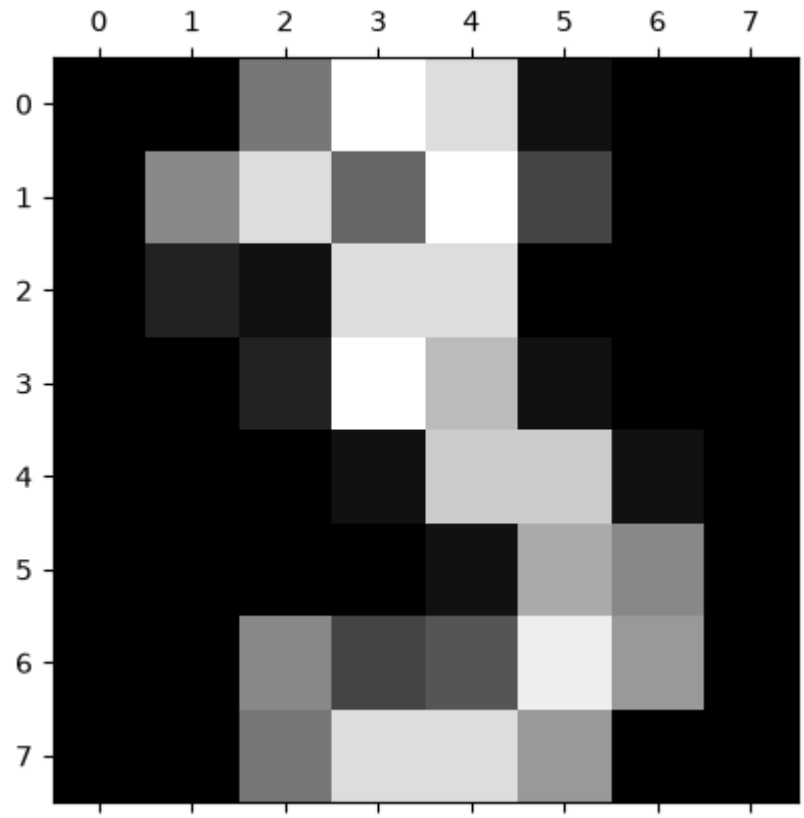
In [5]:
```
%matplotlib inline
plt.gray()
for i in range(4):
    plt.matshow(digits.images[i])
    plt.show()
```

<Figure size 640x480 with 0 Axes>






0 1 2 3 4 5 6 7
0
1
2
3
4
5
6
7


0 1 2 3 4 5 6 7
0
1
2
3
4
5
6
7

In [6]:
```
df = pd.DataFrame(digits.data)
df.head()
```

Out[6]:

	0	1	2	3	4	5	6	7	8	9	...	54	55	56	57	58	59	60
0	0.0	0.0	5.0	13.0	9.0	1.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	6.0	13.0	10.0
1	0.0	0.0	0.0	12.0	13.0	5.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	11.0	16.0
2	0.0	0.0	0.0	4.0	15.0	12.0	0.0	0.0	0.0	0.0	...	5.0	0.0	0.0	0.0	0.0	3.0	11.0
3	0.0	0.0	7.0	15.0	13.0	1.0	0.0	0.0	0.0	8.0	...	9.0	0.0	0.0	0.0	7.0	13.0	13.0
4	0.0	0.0	0.0	1.0	11.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	2.0	16.0

5 rows × 64 columns

In [7]:
```
df['target']= digits.target
df.head()
```

Out[7]:

	0	1	2	3	4	5	6	7	8	9	...	55	56	57	58	59	60	6
0	0.0	0.0	5.0	13.0	9.0	1.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	6.0	13.0	10.0	0.0
1	0.0	0.0	0.0	12.0	13.0	5.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	11.0	16.0	10.0
2	0.0	0.0	0.0	4.0	15.0	12.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	3.0	11.0	16.0
3	0.0	0.0	7.0	15.0	13.0	1.0	0.0	0.0	0.0	8.0	...	0.0	0.0	0.0	7.0	13.0	13.0	9.0
4	0.0	0.0	0.0	1.0	11.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	2.0	16.0	4.0

5 rows × 65 columns

In [8]:
```
from sklearn.model_selection import train_test_split
x_train , x_test , y_train , y_test = train_test_split(df.drop(['target'],axis='c
```

In [9]:
```
len(x_train)
```

Out[9]: 1437

In [10]:
```
len(x_test)
```

Out[10]: 360

In [11]:
```
from sklearn.ensemble import RandomForestClassifier
model = RandomForestClassifier(n_estimators = 100 , criterion = 'gini')
model.fit(x_train , y_train)
```

Out[11]:

RandomForestClassifier

▶ Parameters

▶ Fitted attributes

In [12]:
```
model.score(x_test,y_test)
```

Out[12]: 0.9305555555555556